UNIT 1 · KINEMATICS · WEEK 2 · HONORS

Problem Set 5 (Version 2) — Uniform Acceleration

Physics 2026–27 · due at the start of class, Tuesday, October 6

NAME _____ DATE _____

How to work this version. Same seven problems, same skills, same selection rule — *find the one variable you neither know nor want, and use the equation that omits it* — but the numbers are rougher and most problems add one extra move. Watch the units: some givens arrive in km/h, and one answer is a speed you have to convert back. Name the missing variable in your E row every time. Symbols first, numbers last, units the whole way.

1 Pick the equation — don't solve

SELECTION RULE

For each scenario, name the **missing variable** (neither known nor wanted) and write the equation you'd use. No arithmetic. Two of these hide the missing variable behind a unit conversion or an unusual target — read carefully.

a. A jet starts from rest and accelerates at 3.5 m/s² down 900 m of runway. You want its takeoff speed.

b. A train slows from 144 km/h to 36 km/h over 60 s. You want how far it traveled. (Convert first.)

c. A dropped phone falls 1.6 m. You want its speed just before impact. (Note: you are *not* given the time.)

d. A sprinter accelerates steadily at 2.5 m/s² from 3.0 m/s to 9.0 m/s. You want the distance covered while speeding up.

ANSWERS

2 The rocket sled, already rolling

KDEA

This sled is *not* starting from rest: it is already moving at `8.0 m/s` when the rocket fires, then accelerates at a constant `12 m/s²` for `5.0 s`.

a. Its final velocity.

b. The distance it covers during the burn.

c. How far it *would* have covered in the same 5.0 s if it had started from rest.

d. One sentence: explain the difference between (b) and (c) using the equation, not just the arithmetic.

K	EVERY GIVEN, WITH UNITS
D	THE TARGETS & THEIR UNITS
E	WHICH EQUATION? NAME THE MISSING VARIABLE
A	SYMBOLS FIRST · UNITS THE WHOLE WAY

3 No time given — on purpose

KDEA

A car moving at `22 m/s` brakes with constant acceleration and stops in `40 m`.

a. Find the acceleration (sign included). The word "time" appears nowhere in the problem — let that steer your first E row.

b. Now find how long the braking takes, two ways: once with the kinematics equations, and once with $t = \Delta x / v_{avg}$. Do they agree?

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4 No acceleration given — go deeper

KDEA

A runner speeds up steadily from `2.0 m/s` to `8.0 m/s` over `3.0 s`. Acceleration is constant, but its value never appears.

a. How far does she travel during those 3.0 s?

b. What was her *average* velocity over the interval?

c. One sentence: why is that average exactly the midpoint of 2.0 and 8.0 m/s here — and would it still be the midpoint if the acceleration were not constant?

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5 Three-stage trip

MULTI-STEP

A motorcycle starts from rest, accelerates at `4.0 m/s²` for `6.0 s`, then holds a constant velocity for another `8.0 s`, then brakes uniformly to rest in `5.0 s`.

a. How fast is it moving at the end of the first stage?

b. How far does it travel during the first stage?

c. How far during the constant-velocity stage? (Careful — the four-equation bank isn't even needed here.)

d. What is its acceleration during braking, sign included?

e. Total distance for the whole trip?

WORK

6 Speeding up, slowing down, or reversing?

SIGNS & MOTION

For each part, use the sign rule (same signs vs. opposite signs), and compute a value where one is asked for.

a. $v = +14\ \text{m/s}$, $a = -3\ \text{m/s}^2$. Speeding up or slowing down?

b. An object moving at $-6\ \text{m/s}$ speeds up to $-9\ \text{m/s}$ in $1.5\ \text{s}$. Find its acceleration, then state the motion.

c. $v = +5\ \text{m/s}$, $a = +5\ \text{m/s}^2$. Speeding up or slowing down?

d. An object is moving at $+8\ \text{m/s}$ with $a = -8\ \text{m/s}^2$. Describe its motion over the next 1.0 s — including what happens at $t = 1.0\ \text{s}$ and after.

e. One sentence: why does "negative acceleration" alone tell you nothing about whether the object is slowing down?

WORK

7 Grade this work — two errors

CRITIQUE

The problem: *"A car moving at 30 m/s brakes uniformly to rest in 6.0 s. How far does it travel while stopping?"* A student writes:

$$\Delta x = v_f\, t + \tfrac{1}{2}\, a\, t^2 = (0)(6.0) + \tfrac{1}{2}(-5.0)(36) = -90\ \text{m}$$

a. The value $a = -5.0\ \text{m/s}^2$ is correct. Find **both** things wrong with the student's final line — one is a subscript, one is about what a distance can and cannot be.

b. Fix it: run the full KDEA and report the real stopping distance. Sanity-check the sign of your answer.

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Version 2 is still graded on the homework 4–3–2–1 scale, but the bar is higher: the numbers are deliberately less clean, so a right answer with dishonest rounding or missing units will not earn full marks. Every E row must name the missing variable — that habit is most of this week, in both versions.